FULL STACK DEVELOPMENT

1. HTML for Static Web Pages

Core Concepts

- Introduction is HTML
- Structure of HTML document
- HTML versions (HTML4, XHTML, HTML5)

HTML Building Blocks

- Tags
- Elements
- Attributes
- Empty tags
- Container tags

A. Text & Formatting

- Headings (<h1>–<h6>)
- Paragraphs (<p>)
- Line breaks (<br>)
- Horizontal rule (<hr>)
- Formatting tags
    - <b>, <i>, <u>
    - ,
    - <small>, <mark>
    - <sup>, <sub>

B. Lists

- Ordered List <ol>
  - type attribute
  - start attribute
- Unordered List <ul>
  - type attribute
- List Items <li>
- Nested Lists
- Ordered list inside unordered list
- Definition List
  - <dl>
  - <dt>
  - <dd>

C. Links (Hyperlinks)

- Anchor tag <a>
- href attribute
- target attribute
  - _self
  - _blank
  - _parent
  - _top
- Internal links
- External links,Email links
- Image as link

D. Images

- <img> tag
- src attribute
- alt attribute
- height & width
- Image linking
- Thumbnails
- Image gallery concept

Table Structure

- <table>
- <tr>
- <th>
- <td>
- <caption>

Table Attributes

- border
- cellpadding
- cellspacing
- rowspan
- colspan

Applications

- Simple table
- Timetable
- Data tables
- Forms using tables

Form Basics

- <form>
- action
- method (GET, POST)

Input Controls

- Text field
- Password field
- Number field
- Date field
- Radio buttons
- Checkboxes
- Select box
    - <select>
    - <option>
- Text area <textarea>
- Buttons
    - Submit
    - Reset

HTML FRAMES

- <frameset>
- <frame>
- rows
- cols
- noframes
- Fixed frames
- Navigation using frames

Structural Tags

- <header>
- <footer>
- <main>
- <section>
- <article>
- <aside>
- <nav>

Media Tags

- <audio>
- <video>
- <source>
- controls
- autoplay
- loop

Content Tags

- <figure>
- <figcaption>
-
-

CSS (Cascading Style Sheets)

CSS Fundamentals

- Introduction to CSS
- CSS syntax
    - selector
    - property
    - value

Types of CSS

1. Inline CSS
2. Internal CSS
3. External CSS

Simple Selectors

- Element selector
- ID selector
- Class selector
- Group selector
- Universal selector

Combinator Selectors

- Descendant
- Child
- Adjacent sibling
- General sibling

Pseudo Selectors

- Pseudo-class (:hover, :active, :focus)
- Pseudo-element (::before, ::after, ::first-letter)

Attribute Selectors

- [type]
- [type="text"]
- [href^="https"]

CSS STYLING CONCEPTS

Colors

- Color names
- RGB
- HEX
- HSL
- RGBA

Background

- background-color
- background-image
- background-position
- background-repeat
- background-attachment (fixed)

Fonts

- font-size
- font-weight
- font-style
- font-family

Text

- text-decoration
- text-transform
- text-align
- line-height
- letter-spacing

CSS BOX MODEL

- Content
- Padding
- Border
- Margin
- box-sizing

JavaScript Basics

- Introduction to JavaScript
- Internal JavaScript
- External JavaScript

Input & Output

- alert()
- confirm()
- prompt()
- document.write()
- console.log()
- innerHTML

Type Conversion

- Implicit conversion
- Explicit conversion
- Number()
- String()
- Boolean()

Conditional Statements

- if
- if-else
- else-if
- switch

Loops

- for
- while
- do-while
- for-in
- for-of
- for-each

Predefined Objects

- document object
- window object
- array object
- string object
- math object
- date object
- regex object

User-Defined Objects

- Properties
- Methods
- Constructors
- Accessors (get/set)

Functions

- Function declaration
- Function expression
- Arrow functions

Events

- onclick
- onload
- onchange
- onmouseover
- onsubmit

Applications

- Factorial
- Fibonacci
- Prime numbers
- Palindrome
- Form validation
    - Name validation
    - Mobile validation
    - Email validation

DOM Basics

- What is DOM
- DOM tree
- DOM manipulation

DOM Selection

- getElementById
- getElementsByClassName
- getElementsByTagName
- querySelector
- querySelectorAll

DOM Events

- Event listeners
- Click events
- Form events

NODE.JS (INTRODUCTION)

- What is Node.js
- JavaScript runtime
- Server-side JS
- Uses of Node.js

Core ES6 Concepts

- let & const
- Arrow functions
- Spread operator
- Rest operator
- Template literals
- Destructuring
    - Array destructuring and Object destructuring
- Type coercion and Hoisting

REACT.JS

React Fundamentals

- Introduction to React and Why React
- JSX
- Components
    - Class components
    - Functional components

Virtual DOM

- Real DOM
- Virtual DOM
- Differences
- Reconciliation

React Core Concepts

- Props
- State
- Conditional rendering
- Iterative rendering (map())

React Hooks

- useState
- useEffect

React Forms

- Controlled components
- Input handling
- Form submission

React API Handling

- Fetch API
- API integration
- useEffect with API

GIT & GITHUB

Git Basics

- Version control
- Git installation
- Configuration

Git Commands

- git init
- git status
- git add
- git commit
- git branch
- git checkout
- git merge
- git push
- git pull

GitHub

- Repository creation
- Personal access token
- Remote repo connection
- .gitignore
- Branching strategy
- Pull requests
- Code review
- Issues

NODE.JS & EXPRESS.JS

Node.js

- Runtime environment
- Non-blocking I/O
- Event loop

Express.js

- Express setup
- Routing
- Middleware
- Request & response
- CRUD operations
- MVC architecture
- Microservices architecture

DATABASE – MySQL

MySQL Basics

- Database
- Tables
- Rows & columns

SQL Queries

- CREATE
- INSERT
- UPDATE
- DELETE
- SELECT

Advanced SQL

- Subqueries
- Scripts
- Workbench usage

MySQL with Express

- MySQL driver
- Database connection
- API & DB integration

MongoDB – Basics to CRUD Operations

Introduction to MongoDB

- What is MongoDB?
- NoSQL vs SQL databases
- Features of MongoDB
  - Schema-less
  - Document-oriented
  - High performance
  - Scalability
- Use cases of MongoDB
- Advantages and limitations of MongoDB

MongoDB Architecture

- MongoDB server
- Database
- Collection
- Document
- BSON format (Binary JSON)
- Relationship between:
  - Database → Collection → Document
- Comparison with relational tables

Installation and Setup

- Installing MongoDB Community Edition
- MongoDB Compass (GUI tool)
- MongoDB Shell (mongosh)
- Starting and stopping MongoDB service
- Connecting to MongoDB server

MongoDB Data Model

- Documents
- Fields and values
- Embedded documents
- Arrays
- ObjectId
- Data types in MongoDB:
    - String
    - Number
    - Boolean
    - Array
    - Object
    - Date
    - Null

MongoDB Database Operations

- Show all databases
  show dbs
- Create / switch database
  use database_name
- Current database
  db
- Drop database
  db.dropDatabase()

MongoDB Collection Operations

- Create collection
  db.createCollection()
- View collections
  show collections
- Drop collection
  db.collection.drop()
- Auto-creation of collections

CURU Operations

- Create Operations (Insert)
- Read Operations (Find)
- Update Operations
- Delete Operations